

ECA5601 –Advanced Wireless Communication Systems / 5G / 6G

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Experiment: 8

Steering Angle Variation in MIMO Beamforming Using MATLAB

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Aim:

To analyze the effect of steering angle variation in a MIMO beamforming system by studying steering angle, array gain, and beam direction error using MATLAB.

Objectives:

To analyze the steering angle of a MIMO beamforming system by directing the beam towards different target directions using MATLAB.

Software Required: • MATLAB

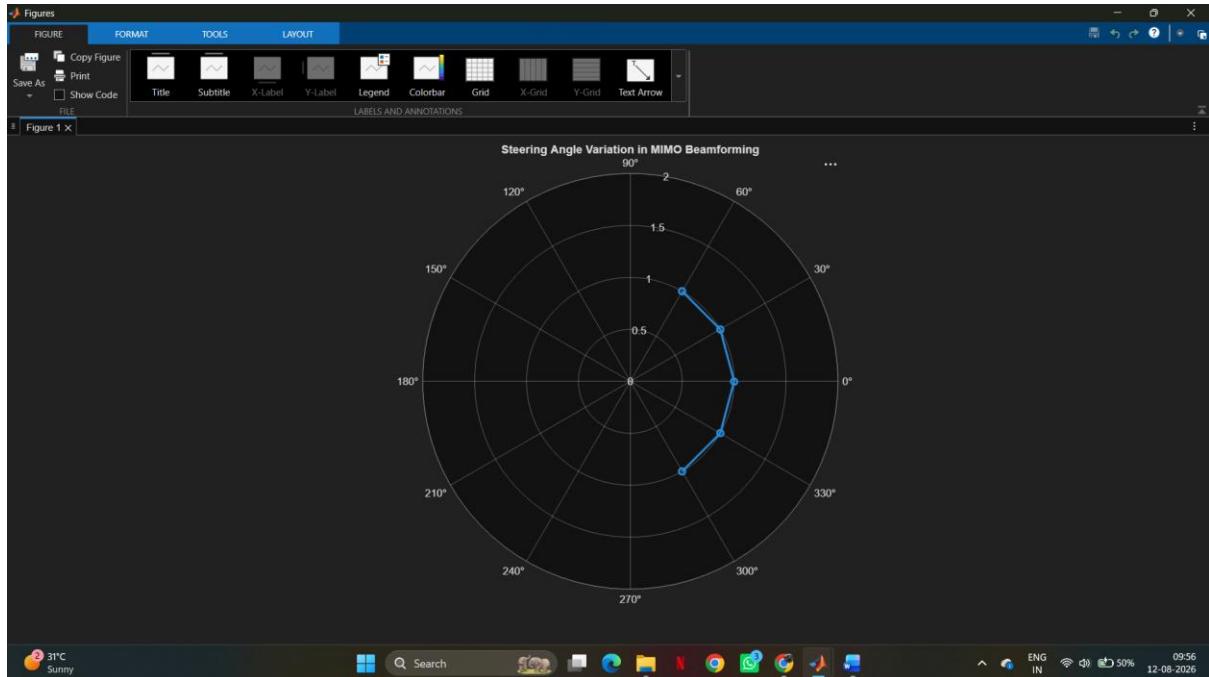
Theory:

MIMO (Multiple-Input Multiple-Output) beamforming enhances wireless communication by employing multiple antenna elements to direct transmitted energy toward a desired user. By adjusting the steering angle through appropriate phase shifts, the beam can be focused in different directions, improving signal strength and reducing interference. Important performance parameters include steering angle, array gain, and beam direction error, which collectively determine the effectiveness of beam steering in modern wireless systems such as 5G NR.

MATLAB Code:

```
clc;
clear;
close all;
angle = [-60 -30 0 30 60]; % Steering Angles (degrees)
gain = ones(size(angle)); % Normalized Gain
figure
polarplot(deg2rad(angle), gain,'o-','LineWidth',2)
title('Steering Angle Variation in MIMO Beamforming')
```

OUTPUT:



Result:

The MATLAB analysis successfully demonstrated steering angle variation in a MIMO beamforming system. The results showed that beam steering effectively directs the transmitted energy toward different angles, while maintaining high array gain and minimizing beam direction error.